

Digital Forensics Worksheet

The Case of the MyFace Attack - Part 1

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Case Summary:

David Eng is a popular Tremble gamer with over 2 million followers. He's employed by the popular video gaming behemoth, SoChi, and he's gained celebrity cult status specifically because of his supposed wholesomeness. On his streams, he doesn't curse or berate others. Instead, he's adopted a "no jerks policy" and puts a positive spin on everything. It's because of this that companies are clamoring for him to endorse their products.

Last month, however, David's professional life took a dramatic turn for the worse. Out of nowhere, salacious pictures from his college days when he participated in his fraternity beer keg stands started appearing on different social media accounts. David was immediately removed from Tremble and he was fired from SoChi shortly thereafter.

A few days ago, David filed a complaint seeking damages from MyFace, the giant social media company. David claims his MyFace account was hacked on the MyFace server. David states that his personal computer was not hacked; it was only his MyFace account. This led directly to those pictures being posted on social media. The reason he is suing is because he lost his job due to the pictures being leaked. In addition, the complaint includes excerpts of purported emails and pictures between David and one of his associates.

Objective:

You are the owner of Digital Hound Dogs Investigations, a digital forensic detective agency, and you are contracted by MyFace and the owner of MyFace, Ms. Puckerbarge. You have been hired to review David Eng's Facebook cloud attack.

Pursuant to the court order, you have collected digital media made available by David Eng. You inspected the data on the media/cloud for analysis according to the terms of the court-ordered protocol, which states that you can search and analyze the evidence "to identify only documents, data, fragments, and artifacts that reasonably appear[ed] to be related to the authenticity of the [Work for Hire Document] attached to the Amended Complaint and the purported emails described in the Amended Complaint."

You have also collected evidence from one of the MyFace servers in David's region.

By the end of Week 5, you will complete a Digital Forensics Evidence Report with your findings for MyFace so they can use it in court.

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For Weeks 1 through 4, you will need to complete a series of Digital Forensics Worksheets.

Digital Evidence:

- MyFace server or servers in David's region.
 - MyFace account user login information
 - *Tip!* Look for non-repudiation such as user account authorization, authentication, and privileges.
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Instructions:

To complete this Week 1 worksheet, you'll need to write a 1-page summary on the tools you would use to analyze the evidence presented above.

Step 1: Review the digital evidence listed above.

Step 2: Research tools used by forensics analysts to analyze that kind of evidence.

Here are some suggested software tools to research which may help you in your investigation. You do NOT need to use these. You can choose your own.

- CloudStrike
- Hype Cycle
- SourceForge

Step 3: Choose two tools that you would use to analyze the evidence.

Step 4: Write a summary of how you would use these tools to analyze the evidence presented.

In your 1-page summary:

- Write a structured 1–2-page summary explaining how you would use selected forensic tools to analyze the evidence presented in David Eng versus MyFace.
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- In your summary:
 - Select two appropriate digital forensic tools that would support the analysis of the evidence presented in the case.
 - Justify your selection by explaining how each tool functions within a digital investigation and how it supports evidence identification, preservation, validation, or analysis.
 - Examine how human behavior, insider threats, and social engineering may have influenced the incident and discuss how these behavioral factors shape investigative strategy within criminal justice, cybersecurity, and homeland security environments.
 - Demonstrate professional clarity, structured reasoning, and appropriate use of digital forensic terminology throughout your response.

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- Your analysis should reflect both technical investigative procedures and the broader operational context in which digital investigations occur.
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- Make sure your writing is at an appropriate level and clarity.

Summary:

As the owner of Digital Hound Dogs Investigations, I was contracted by MyFace to investigate the alleged compromise of David Eng's MyFace account. The objective of the investigation is to determine whether David's account was hacked through MyFace systems, identify evidence related to account access and authentication, and validate the authenticity of the relevant emails and digital artifacts. To conduct this investigation, I would use Magnet AXIOM and Splunk Enterprise Security, as both tools offer strong capabilities for collecting, preserving, validating, and analyzing digital evidence from cloud environments and server logs.

The first tool I chose, Magnet AXIOM, is a widely used digital forensics platform that collects and analyzes evidence from computers, cloud accounts, mobile devices, and social media platforms. In this case, Magnet AXIOM would be used to examine David Eng's MyFace account data, login records, account activity, and cloud-based artifacts. The tool can recover deleted data, analyze metadata, and identify timestamps associated with account access. This information would help determine whether unauthorized access occurred and whether the leaked photographs originated from David's account. Magnet AXIOM also maintains forensic integrity by documenting evidence of acquisition and generating reports that support chain-of-custody requirements. These capabilities comply with evidence preservation, validation, and presentation in court.

The second tool, Splunk Enterprise Security, would be used to analyze server logs collected from MyFace servers in David's geographic region. Splunk is designed to aggregate

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and correlate large amounts of security data, making it effective for identifying suspicious login activity, unusual authentication attempts, IP addresses, and account privilege changes. Since the case focuses on whether David's MyFace account was compromised, Splunk can help determine if unauthorized users accessed the account, whether login attempts originated from unfamiliar locations, and if any indicators of compromise are visible in the server logs. Splunk also supports non-repudiation by providing detailed authentication and authorization records that can verify who accessed the account and when the access occurred.

Human behavior must also be considered during the investigation. Social engineering attacks, such as phishing emails or credential theft schemes, are common methods used to gain access to online accounts. Even if MyFace's systems were not directly breached, an attacker may have manipulated David or an associate into revealing account credentials. Insider threats must also be considered because individuals with legitimate access to sensitive information may intentionally or unintentionally expose data. Understanding these behavioral factors helps investigators develop a broader investigative strategy and avoid focusing only on technical evidence.

Within criminal justice, cybersecurity, and homeland security environments, successful investigations require both technical analysis and an understanding of human behavior. By combining Magnet AXIOM's forensic analysis capabilities with Splunk's log correlation and authentication monitoring features, investigators can conduct a strong examination of the evidence while maintaining ethical standards and accountability. These tools provide the necessary support to identify, preserve, validate, and analyze evidence relevant to the claims made in David Eng's lawsuit against MyFace.

Sources:

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Nelson, B., Phillips, A., & Steuart, C. (2025). *Guide to computer forensics and investigations* (7th ed.). Cengage Learning.

Magnet Forensics. (2025). *Magnet AXIOM*. <https://www.magnetforensics.com/products/magnet-axiom/>

Splunk Inc. (2025). *Splunk Enterprise Security*.
https://www.splunk.com/en_us/products/enterprise-security.html

This summary was written by me and edited with the help of Grammarly to maintain a professional tone.